LAB 7 – Fuzz Testing Report

Bowling Alley Input Validation

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Assigned project	Bowling Alley | Repository 64

Objective

Use Hypothesis property-based fuzz testing to expose failures caused by unexpected input in the bowling alley name, frame, and scoreboard functions, then harden the functions and rerun the tests.

Tested functions

format_player_name, parse_frame, and parse_scoreboard

Result

The buggy implementation failed the three fuzz properties while the valid-example checks passed. After defensive input handling was added, the test suite passed all four tests. The corrected SS4 terminal screenshot is reserved below and will be inserted when supplied.

Assigned repository: https://github.com/AhanaSrinivas/L6_L7_64_bowling_alley

SS1 Buggy application output

The application accepts representative valid inputs before fuzz testing.

```
PS D:\Desktop\PES\SEM5\SE\Lab n project\LABS\LAB 7> python processor.py
==== Running format_player_name ====
Enter a player name for the scoreboard: Asha Patel
Scoreboard name: Asha Patel

==== Running parse_frame ====
Enter a frame as X (strike), 7/ (spare) or two digits (e.g. 45): 7/
Pins (first roll, second roll): (7, 3)

==== Running parse_scoreboard ====
Enter scores as NAME:SCORE separated by commas (e.g. Asha:150,Ravi:132): Asha:150,Ravi:132
Scoreboard: {'Asha': 150, 'Ravi': 132}
```

Figure 1. Starter application output using Asha Patel, 7/, and Asha:150,Ravi:132.

SS2 Hypothesis failures on the buggy implementation

Hypothesis minimized invalid inputs and exposed crashes in each of the three functions. The terminal summary reports three failed tests and one passing valid-example test.

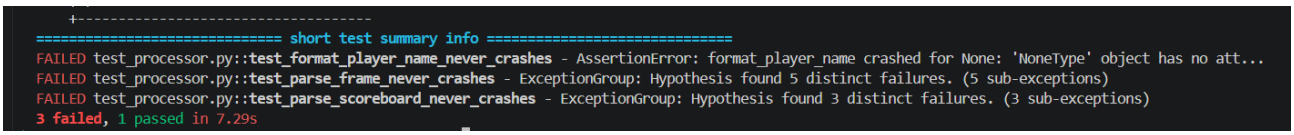


Figure 2. Failure summary from the buggy implementation.

SS3 Fixed application output

The same representative valid inputs continue to produce the expected output after input validation was added.

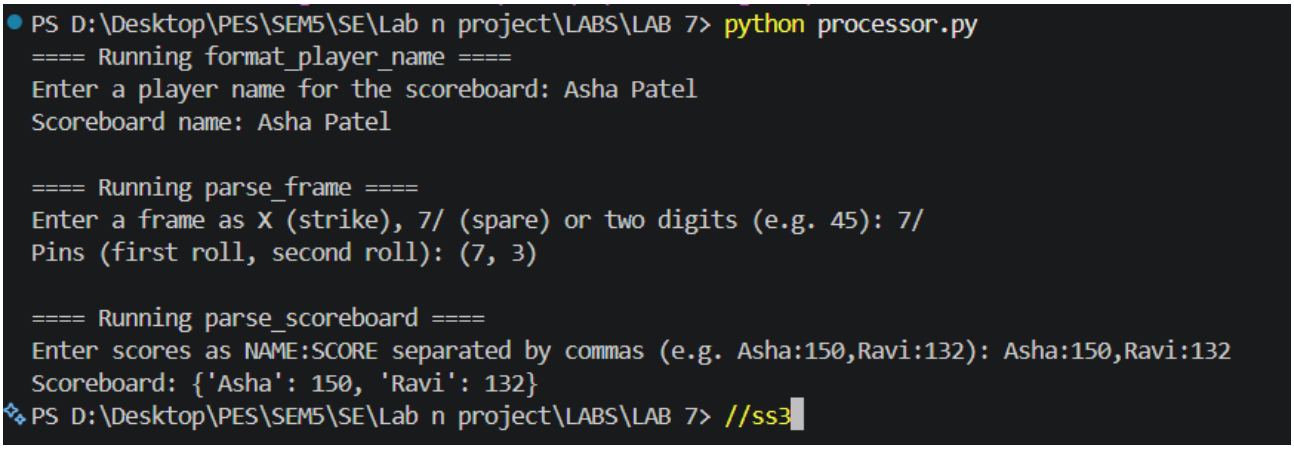
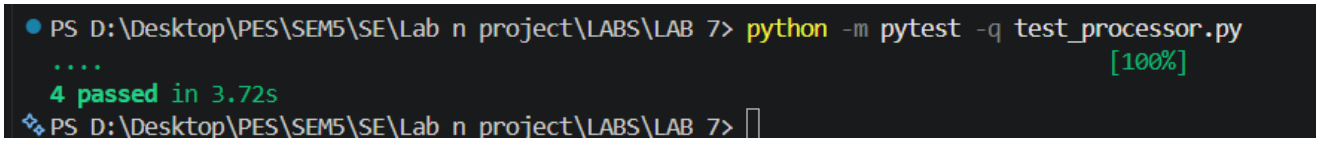


Figure 3. Fixed application output using the same valid inputs as SS1.

SS4 Passing Hypothesis tests

```
python -m pytest -q test_processor.py
```



Additional SS2 failure details

The following screenshots continue the terminal traceback for SS2 and are included as supplementary evidence.

SS2 failure details 1 of 4

```
PS D:\Desktop\PES\SEM5\SE\Lab n project\LABS\LAB 7> python -m pytest -q test_processor.py
FFF.                                                                     [100%]
================================== FAILURES ===================================
_____________________ test_format_player_name_never_crashes _____________________

value = None

    @given(st.one_of(st.text(), st.none(), st.integers(), st.lists(st.text())))
    def test_format_player_name_never_crashes(value):
        try:
>           result = format_player_name(value)
                     ^^^^^^^^^^^^^^^^^^^^^^^^^

test_processor.py:11:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

name = None

    def format_player_name(name):
        """
        Title-cases a player name and cuts it to 10 characters for the scoreboard.
        Assumes name is a non-empty string.
        """
>       return name.strip().title()[:10]
               ^^^^^^^^^^
E       AttributeError: 'NoneType' object has no attribute 'strip'

processor.py:6: AttributeError

During handling of the above exception, another exception occurred:

    @given(st.one_of(st.text(), st.none(), st.integers(), st.lists(st.text())))
>   def test_format_player_name_never_crashes(value):
                ^^^

test_processor.py:9:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

value = None

    @given(st.one_of(st.text(), st.none(), st.integers(), st.lists(st.text())))
    def test_format_player_name_never_crashes(value):
        try:
            result = format_player_name(value)
        except Exception as exc:
>           assert False, f"format_player_name crashed for {value!r}: {exc}"
E           AssertionError: format_player_name crashed for None: 'NoneType' object has no attribute 'strip'
E           assert False
E           Failing test case: test_format_player_name_never_crashes(
E               value=None,
E           )
E           Explanation:
E               These lines were always and only run by failing test cases:
E                   D:\Desktop\PES\SEM5\SE\Lab n project\LABS\LAB 7\test_processor.py:12

test_processor.py:13: AssertionError
```

SS2 failure details 2 of 4

```
test_processor.py:13: AssertionError
_____________________ test_parse_frame_never_crashes _____________________
  + Exception Group Traceback (most recent call last):
  |   File "D:\Desktop\PES\SEM5\SE\Lab n project\LABS\LAB 7\test_processor.py", line 19, in test_parse_frame_never_crashes
  |     def test_parse_frame_never_crashes(value):
  |                ^^^
  |   File "C:\Users\ahana\AppData\Roaming\Python\Python313\site-packages\hypothesis\core.py", line 2334, in wrapped_test
  |     raise the_error_hypothesis_found
  | ExceptionGroup: Hypothesis found 5 distinct failures. (5 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "D:\Desktop\PES\SEM5\SE\Lab n project\LABS\LAB 7\test_processor.py", line 21, in test_parse_frame_never_crashes
    |     result = parse_frame(value)
    |   File "D:\Desktop\PES\SEM5\SE\Lab n project\LABS\LAB 7\processor.py", line 20, in parse_frame
    |     return first, int(text[1])
    |                   ~~~^^^^^^^^^
    | ValueError: invalid literal for int() with base 10: ':'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "D:\Desktop\PES\SEM5\SE\Lab n project\LABS\LAB 7\test_processor.py", line 23, in test_parse_frame_never_crashes
    |     assert False, f"parse_frame crashed for {value!r}: {exc}"
    | AssertionError: parse_frame crashed for '0:': invalid literal for int() with base 10: ':'
    | assert False
    | Failing test case: test_parse_frame_never_crashes(
    |     value='0:',
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         D:\Desktop\PES\SEM5\SE\Lab n project\LABS\LAB 7\test_processor.py:22
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "D:\Desktop\PES\SEM5\SE\Lab n project\LABS\LAB 7\test_processor.py", line 21, in test_parse_frame_never_crashes
    |     result = parse_frame(value)
    |   File "D:\Desktop\PES\SEM5\SE\Lab n project\LABS\LAB 7\processor.py", line 17, in parse_frame
    |     first = int(text[0])
    | ValueError: invalid literal for int() with base 10: ':'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "D:\Desktop\PES\SEM5\SE\Lab n project\LABS\LAB 7\test_processor.py", line 23, in test_parse_frame_never_crashes
    |     assert False, f"parse_frame crashed for {value!r}: {exc}"
    | AssertionError: parse_frame crashed for ':': invalid literal for int() with base 10: ':'
    | assert False
    | Failing test case: test_parse_frame_never_crashes(
    |     value=':',  # or any other generated value
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         D:\Desktop\PES\SEM5\SE\Lab n project\LABS\LAB 7\test_processor.py:22
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "D:\Desktop\PES\SEM5\SE\Lab n project\LABS\LAB 7\test_processor.py", line 21, in test_parse_frame_never_crashes
    |     result = parse_frame(value)
    |   File "D:\Desktop\PES\SEM5\SE\Lab n project\LABS\LAB 7\processor.py", line 18, in parse_frame
    |     if text[1] == '/':
    |        ~~~~^^^
    | IndexError: string index out of range
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
```

SS2 failure details 3 of 4

```
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "D:\Desktop\PES\SEM5\SE\Lab n project\LABS\LAB 7\test_processor.py", line 23, in test_parse_frame_never_crashes
    |     assert False, f"parse_frame crashed for {value!r}: {exc}"
    | AssertionError: parse_frame crashed for '0': string index out of range
    | assert False
    | Failing test case: test_parse_frame_never_crashes(
    |     value='0',
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         D:\Desktop\PES\SEM5\SE\Lab n project\LABS\LAB 7\test_processor.py:22
    +---------------- 4 ----------------
    | Traceback (most recent call last):
    |   File "D:\Desktop\PES\SEM5\SE\Lab n project\LABS\LAB 7\test_processor.py", line 21, in test_parse_frame_never_crashes
    |     result = parse_frame(value)
    |   File "D:\Desktop\PES\SEM5\SE\Lab n project\LABS\LAB 7\processor.py", line 17, in parse_frame
    |     first = int(text[0])
    |             ~~~~^^^
    | IndexError: string index out of range
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "D:\Desktop\PES\SEM5\SE\Lab n project\LABS\LAB 7\test_processor.py", line 23, in test_parse_frame_never_crashes
    |     assert False, f"parse_frame crashed for {value!r}: {exc}"
    | AssertionError: parse_frame crashed for '': string index out of range
    | assert False
    | Failing test case: test_parse_frame_never_crashes(
    |     value='',
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         D:\Desktop\PES\SEM5\SE\Lab n project\LABS\LAB 7\test_processor.py:22
    +---------------- 5 ----------------
    | Traceback (most recent call last):
    |   File "D:\Desktop\PES\SEM5\SE\Lab n project\LABS\LAB 7\test_processor.py", line 21, in test_parse_frame_never_crashes
    |     result = parse_frame(value)
    |   File "D:\Desktop\PES\SEM5\SE\Lab n project\LABS\LAB 7\processor.py", line 14, in parse_frame
    |     text = text.strip().upper()
    |            ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'strip'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "D:\Desktop\PES\SEM5\SE\Lab n project\LABS\LAB 7\test_processor.py", line 23, in test_parse_frame_never_crashes
    |     assert False, f"parse_frame crashed for {value!r}: {exc}"
    | AssertionError: parse_frame crashed for None: 'NoneType' object has no attribute 'strip'
    | assert False
    | Failing test case: test_parse_frame_never_crashes(
    |     value=None,
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         D:\Desktop\PES\SEM5\SE\Lab n project\LABS\LAB 7\test_processor.py:22
    +-----------------------------------
_____________________ test_parse_scoreboard_never_crashes _____________________
  + Exception Group Traceback (most recent call last):
  |   File "D:\Desktop\PES\SEM5\SE\Lab n project\LABS\LAB 7\test_processor.py", line 32, in test_parse_scoreboard_never_crashes
  |     def test_parse_scoreboard_never_crashes(value):
  |                    ^^^
  |   File "C:\Users\ahana\AppData\Roaming\Python\Python313\site-packages\hypothesis\core.py", line 2334, in wrapped_test
```

SS2 failure details 4 of 4



```
_____ test_parse_scoreboard_never_crashes _____
  + Exception Group Traceback (most recent call last):
  |   File "D:\Desktop\PES\SEM5\SE\Lab n project\LABS\LAB 7\test_processor.py", line 32, in test_parse_scoreboard_never_crashes
  |     def test_parse_scoreboard_never_crashes(value):
  |                ^^^
  |   File "C:\Users\ahana\AppData\Roaming\Python\Python313\site-packages\hypothesis\core.py", line 2334, in wrapped_test
  |     raise the_error_hypothesis_found
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "D:\Desktop\PES\SEM5\SE\Lab n project\LABS\LAB 7\test_processor.py", line 34, in test_parse_scoreboard_never_crashes
    |     result = parse_scoreboard(value)
    |   File "D:\Desktop\PES\SEM5\SE\Lab n project\LABS\LAB 7\processor.py", line 30, in parse_scoreboard
    |     scores[name.strip()] = int(score)
    |                            ~~~^^^^^^^
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "D:\Desktop\PES\SEM5\SE\Lab n project\LABS\LAB 7\test_processor.py", line 36, in test_parse_scoreboard_never_crashes
    |     assert False, f"parse_scoreboard crashed for {value!r}: {exc}"
    | AssertionError: parse_scoreboard crashed for ':': invalid literal for int() with base 10: ''
    | assert False
    | Failing test case: test_parse_scoreboard_never_crashes(
    |     value=':',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "D:\Desktop\PES\SEM5\SE\Lab n project\LABS\LAB 7\test_processor.py", line 34, in test_parse_scoreboard_never_crashes
    |     result = parse_scoreboard(value)
    |   File "D:\Desktop\PES\SEM5\SE\Lab n project\LABS\LAB 7\processor.py", line 29, in parse_scoreboard
    |     name, score = entry.split(':')
    |     ^^^^^^^^^^^
    | ValueError: not enough values to unpack (expected 2, got 1)
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "D:\Desktop\PES\SEM5\SE\Lab n project\LABS\LAB 7\test_processor.py", line 36, in test_parse_scoreboard_never_crashes
    |     assert False, f"parse_scoreboard crashed for {value!r}: {exc}"
    | AssertionError: parse_scoreboard crashed for '': not enough values to unpack (expected 2, got 1)
    | assert False
    | Failing test case: test_parse_scoreboard_never_crashes(
    |     value='',
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "D:\Desktop\PES\SEM5\SE\Lab n project\LABS\LAB 7\test_processor.py", line 34, in test_parse_scoreboard_never_crashes
    |     result = parse_scoreboard(value)
    |   File "D:\Desktop\PES\SEM5\SE\Lab n project\LABS\LAB 7\processor.py", line 28, in parse_scoreboard
    |     for entry in text.split(','):
    |                  ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "D:\Desktop\PES\SEM5\SE\Lab n project\LABS\LAB 7\test_processor.py", line 36, in test_parse_scoreboard_never_crashes
    |     assert False, f"parse_scoreboard crashed for {value!r}: {exc}"
    | AssertionError: parse_scoreboard crashed for None: 'NoneType' object has no attribute 'split'
    | assert False
    | Failing test case: test_parse_scoreboard_never_crashes(
    |     value=None,  # or any other generated value
```